Vivekananda College of Engineering & Technology
[A Unit of Vivekananda Vidyavardhaka Sangha Puttur®]
Affiliated to Visvesvaraya Technological University
Approved by AICTE New Delhi & Recognised by Govt of Karnataka
DEPARTMENT OF CSE (DATA SCIENCE)

Project Phase I Presentation on

Web-Based Automatic Timetable Scheduler for Colleges

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➢ Introduction

Timetable scheduling is a core administrative task in educational institutions that involves managing multiple constraints such as faculty availability, course requirements, and class distribution. Traditionally, this process is carried out manually, which is time-consuming and prone to errors. With increasing academic complexity, the need for an automated system capable of efficiently generating accurate and conflict-free timetables has become evident. This requirement is addressed by the Web-Based Automatic Timetable Scheduler, which provides a comprehensive web-based solution for automated timetable generation..

➢ Objectives

- To automate timetable creation and eliminate inefficiencies and errors caused by manual scheduling.
- To use a Genetic Algorithm to optimize course and faculty allocation while satisfying hard constraints and minimizing soft constraint violations.
- To provide a user-friendly interface for easy data input, schedule modification, and timetable visualization.
- To ensure accurate and reliable timetable generation through validation checks and conflict detection mechanisms.

➢ Problem Statement

Timetable scheduling is an essential administrative activity in educational institutions that involves coordinating faculty, courses, and class schedules. As academic structures become more complex, effective scheduling plays a crucial role in ensuring smooth institutional functioning. To support this requirement, the Web-Based Automatic Timetable Scheduler provides a centralized and automated platform for generating academic timetables in an efficient and organized manner.

➢ Proposed Solution

The proposed solution is a Web-Based Automatic Timetable Scheduler designed to automate the process of academic timetable generation. The system uses a Genetic Algorithm to produce optimized and conflict-free timetables based on predefined academic constraints such as faculty availability, course requirements, and lab allocations. Through a user-friendly web interface, administrators can easily manage academic data and generate timetables efficiently. The solution minimizes manual effort, improves scheduling accuracy, and provides flexibility to handle institutional scheduling needs.

➢ Methodology

The methodology followed here begins with collecting academic data such as departments, courses, faculty details, and availability preferences through a web-based interface. This data is processed using a Genetic Algorithm, which generates multiple timetable solutions and iteratively improves them by minimizing conflicts and satisfying scheduling constraints. The best solution is selected based on fitness evaluation and presented as the final timetable, ensuring efficiency and accuracy in scheduling.

➢ System Architecture

- The Architecture follows 3-layered modular approach.
- User Interface is developed using Streamlit used for form-based input and data visualization.
- The Genetic Algorithm serves as the core optimization engine, generating conflict-free timetables by evaluating multiple scheduling solutions. Genetic Algorithm parameters such as population size, crossover rate, mutation rate, and number of generations control the optimization process and influence the quality of the generated timetable.
- Academic and scheduling data is stored and managed using a database.
- The final optimized timetable is generated and displayed to the user through the web interface.

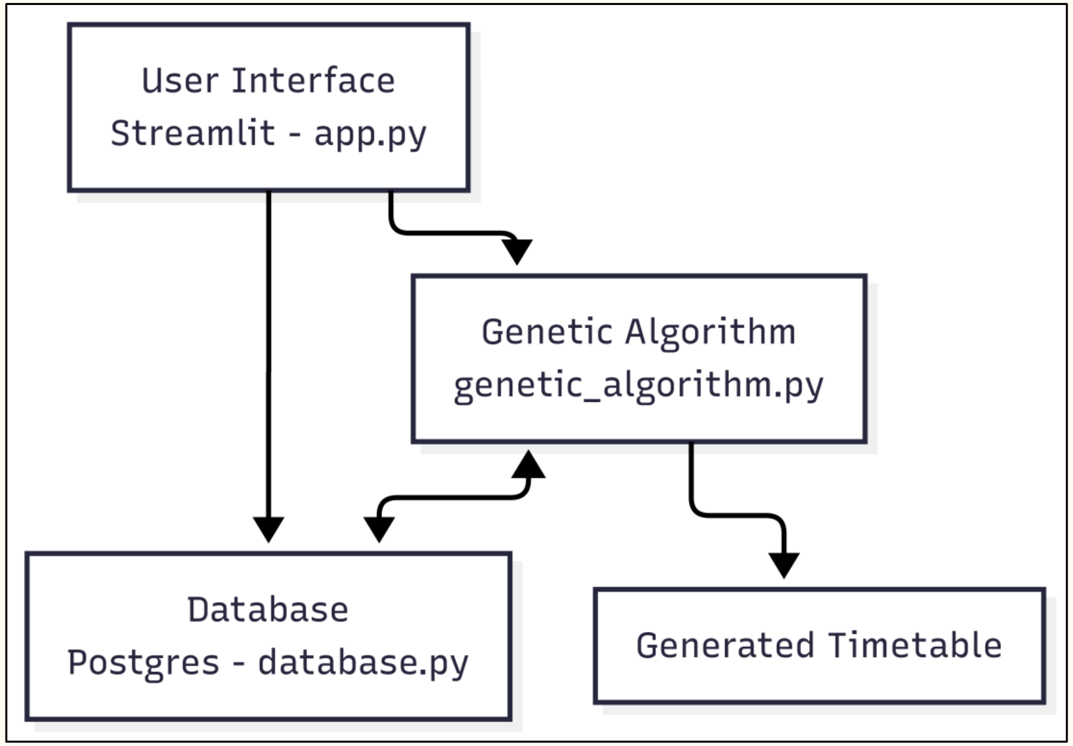


Figure : The System Architecture of Web-based Automatic Timetable Scheduler

➢ Flow Chart

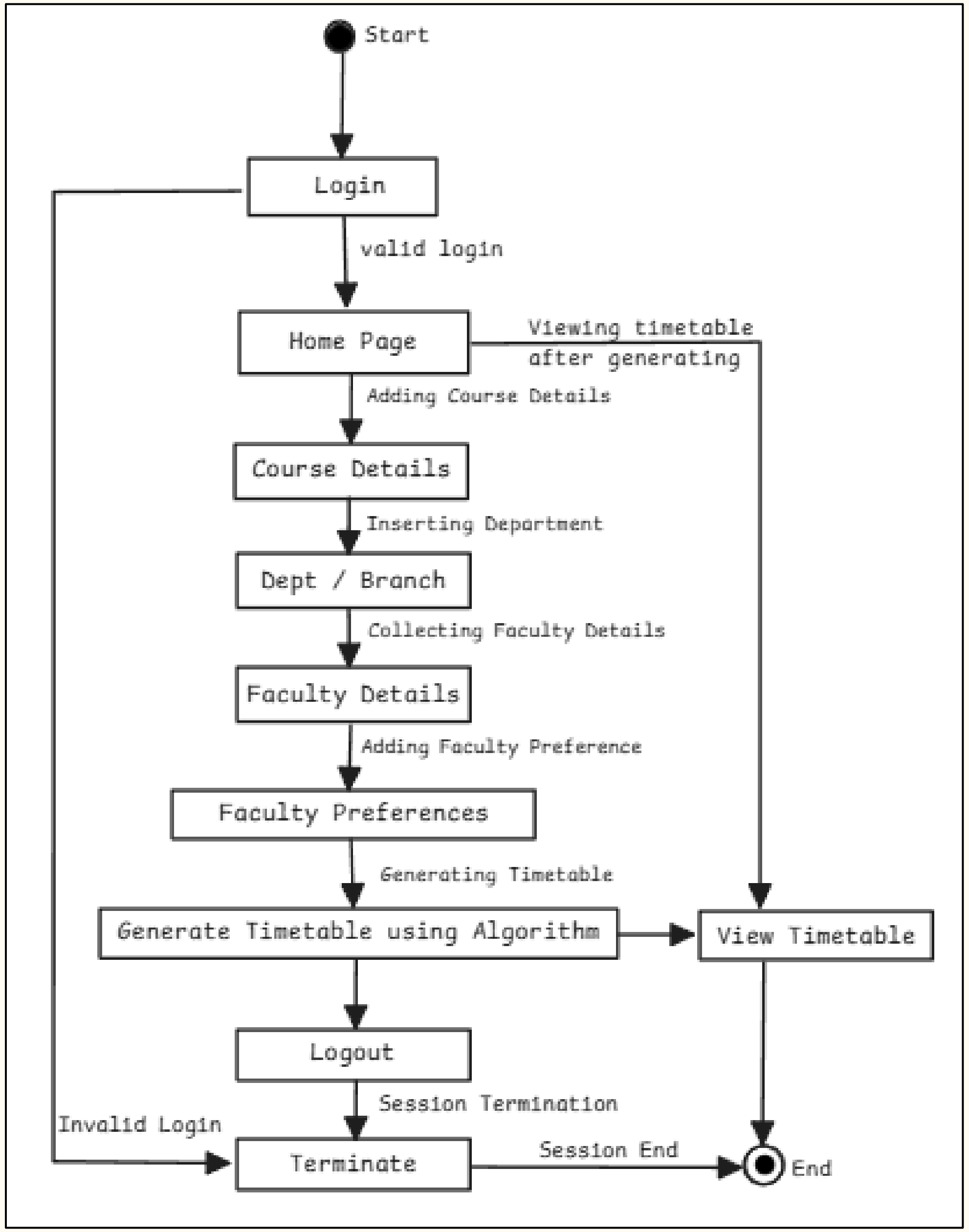


Figure: The system flow diagram of Web-Based Automatic timetable scheduler

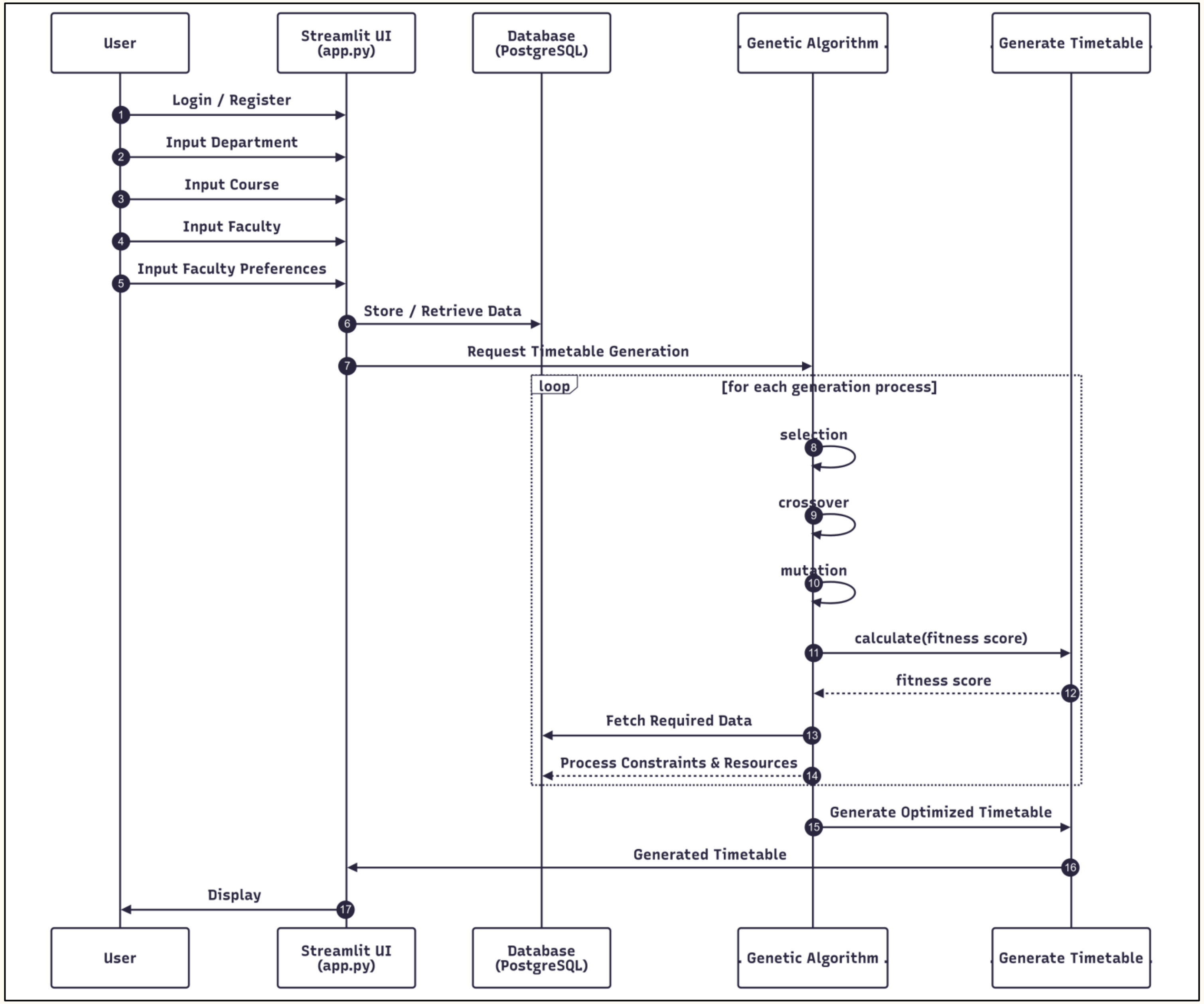
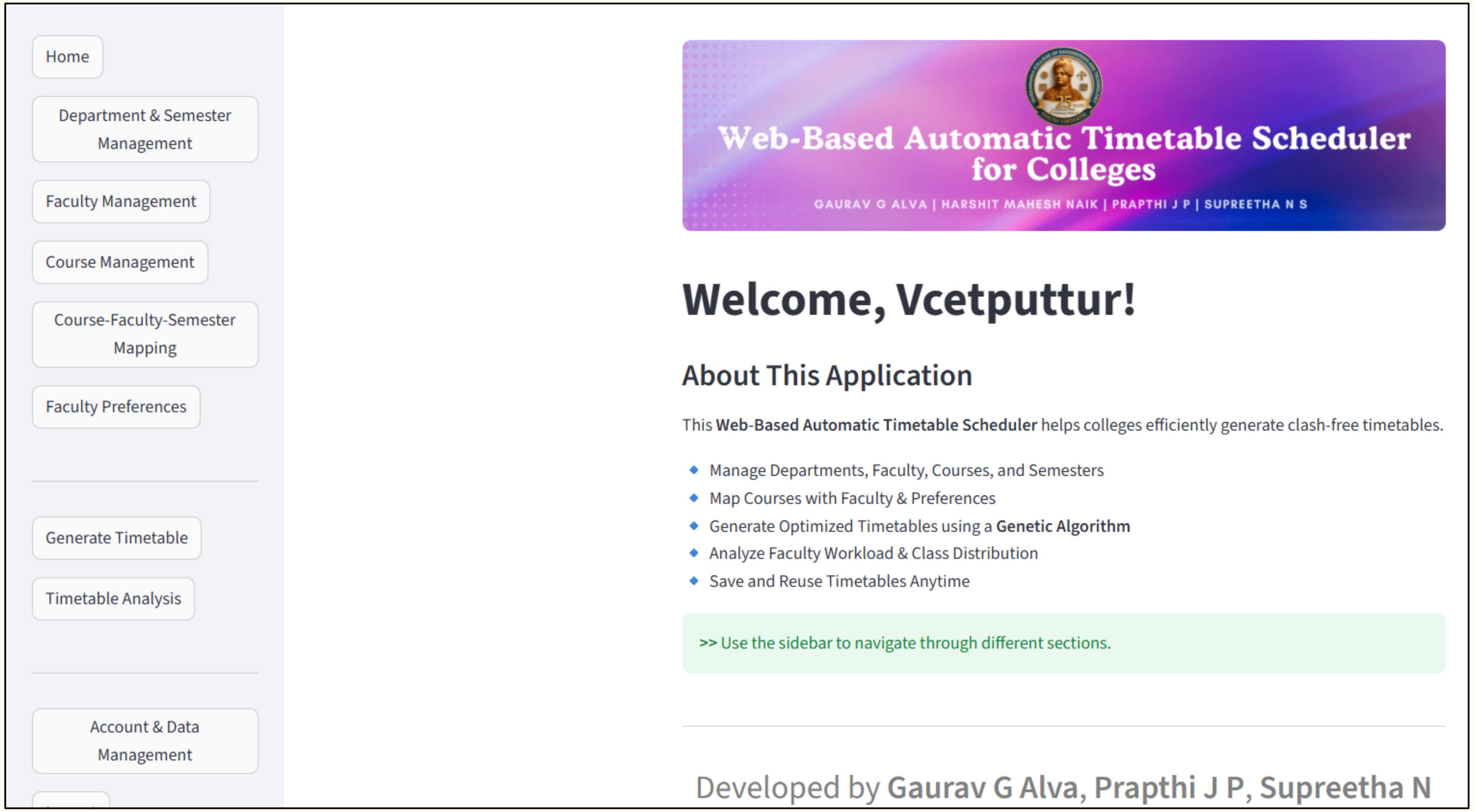


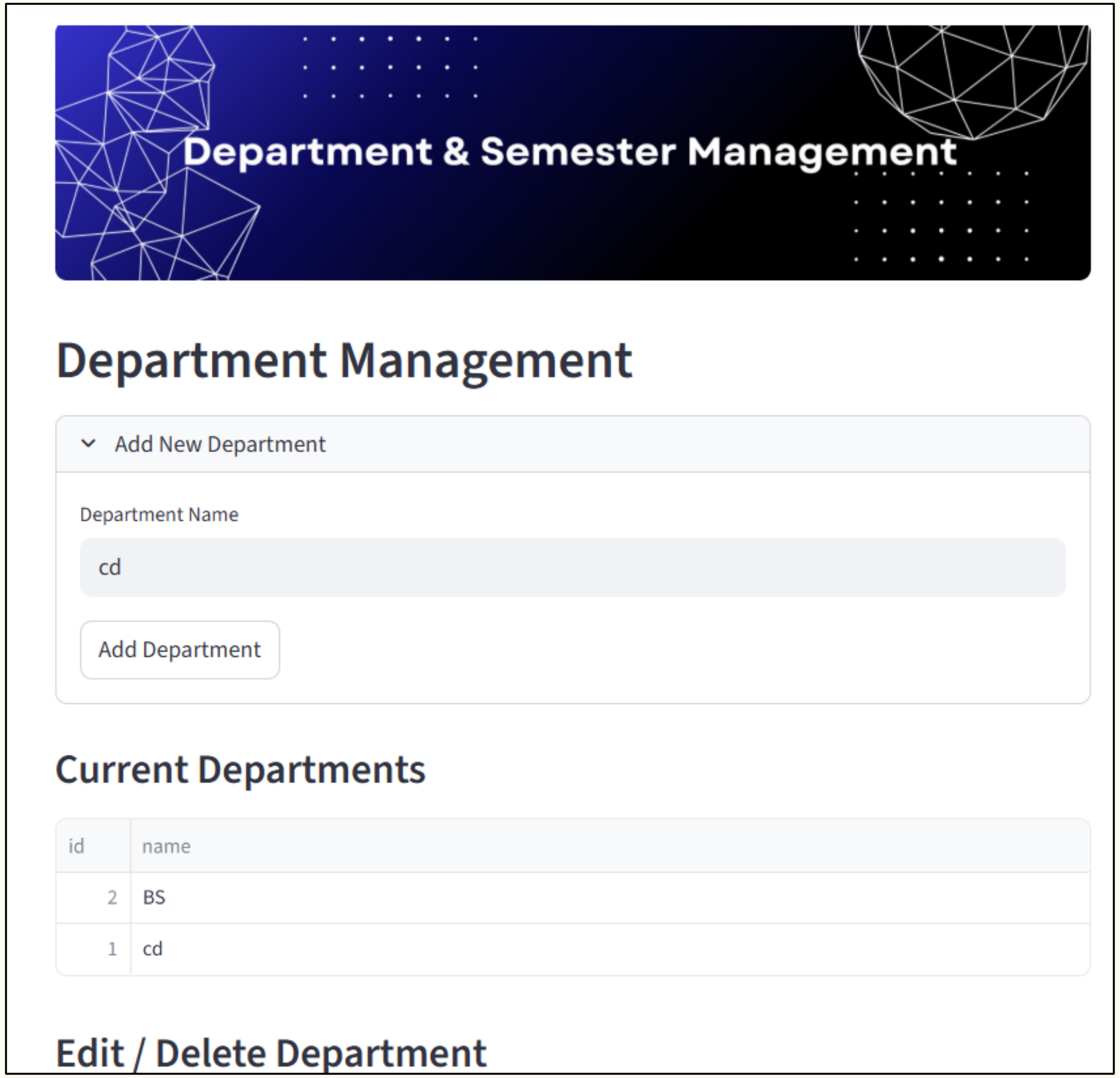
Figure : The activity diagram provides a detailed workflow of the web-based timetable scheduling system, focusing on user interactions and system functions.

➢ Screenshots

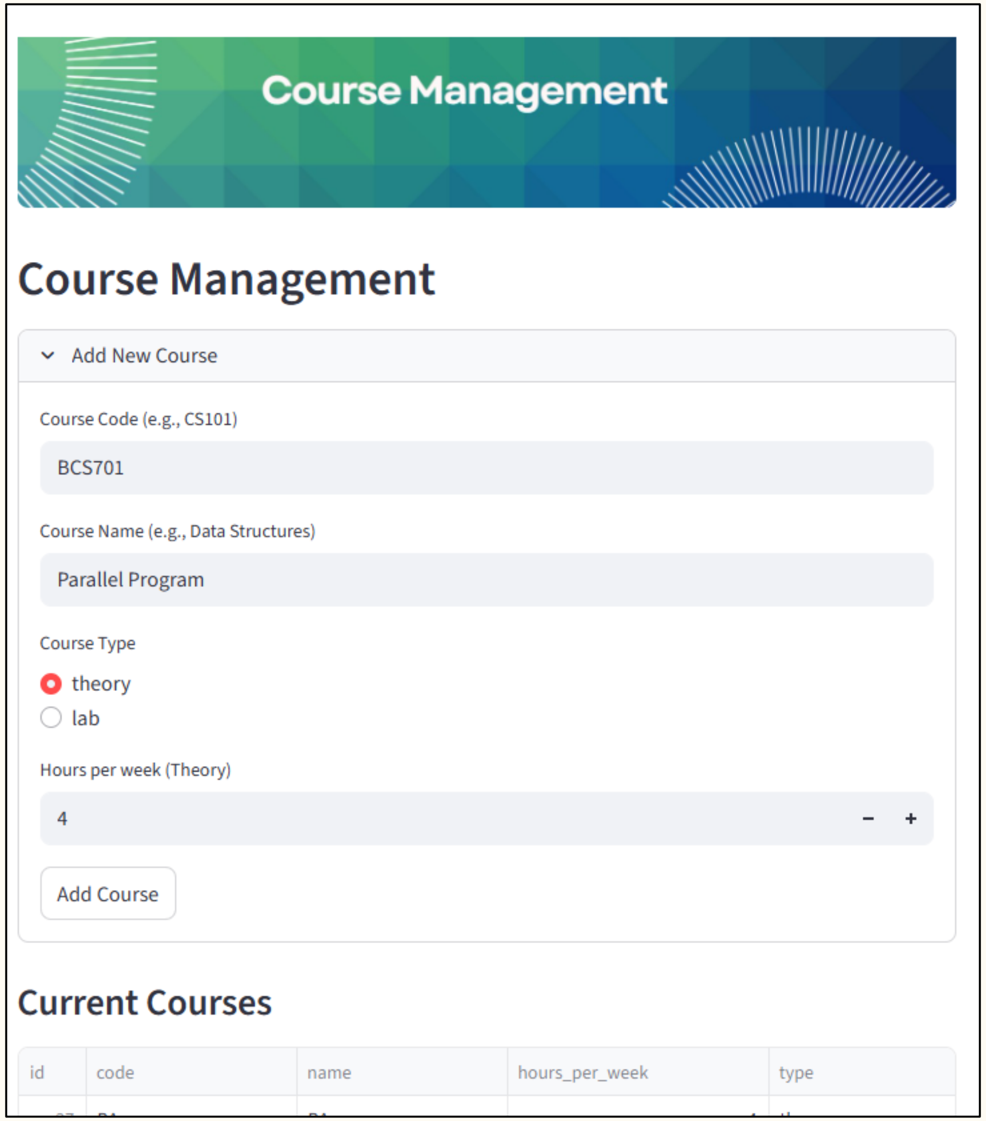


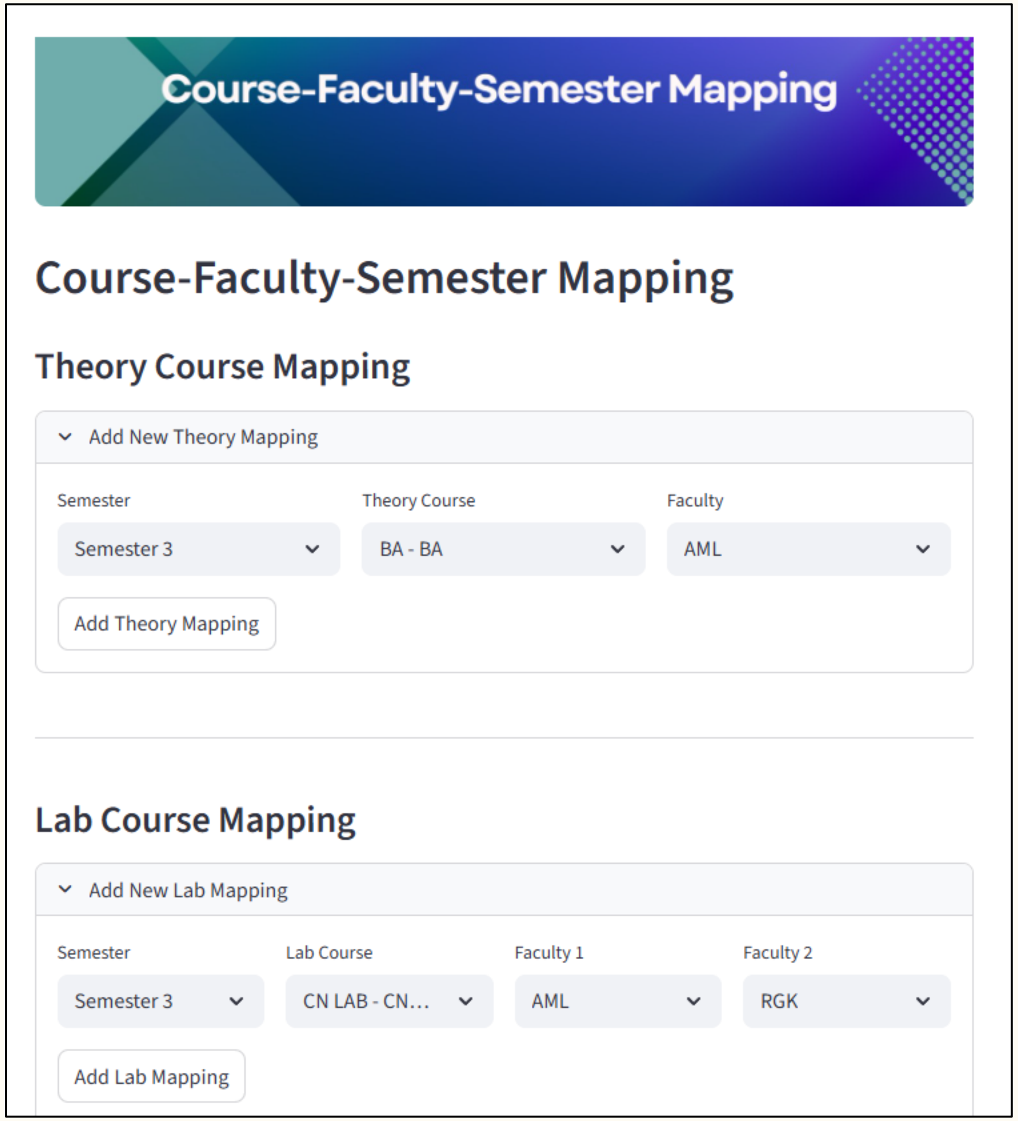
Home Page with the Instruction for timetable generation

Department, Semester and faculty management page



Course management and Course faculty Sem mapping page





Generated Timetable and Downloaded Timetable

Timetable by Semester

Semester 3 Timetable

	Period 1	Period 2	Period 3	Period 4	Period 5	Period 6
Monday	MATH (SMM)		DSA (RHP)	MATH (SMM)	DDCO LAB (AML, NNS)	DDCO LAB (AML, NNS)
Tuesday	MATH (SMM)	DDCO (AML)	OS (SVP)	DSA (RHP)	DA LAB (SRM, CD)	DA LAB (SRM, CD)
Wednesday		OS (SVP)		MATH (SMM)		
Thursday	OS (SVP)	DSA (RHP)	PPDS (SKN)	SCR (SKN)	OS LAB (SVP, SKN)	OS LAB (SVP, SKN)
Friday	NSS (AML)	DSA (RHP)	PE (RHP)	PPDS (SKN)	DSA LAB (RHP, RGK)	DSA LAB (RHP, RGK)
Saturday	DDCO (AML)		DDCO (AML)	PPDS (SKN)	PPDS LAB (SVP, SRM)	PPDS LAB (SVP, SRM)

Semester 5 Timetable

	Period 1	Period 2	Period 3	Period 4	Period 5	Period 6
Monday	CN	TC	RMI		DV LAB	DV LAB

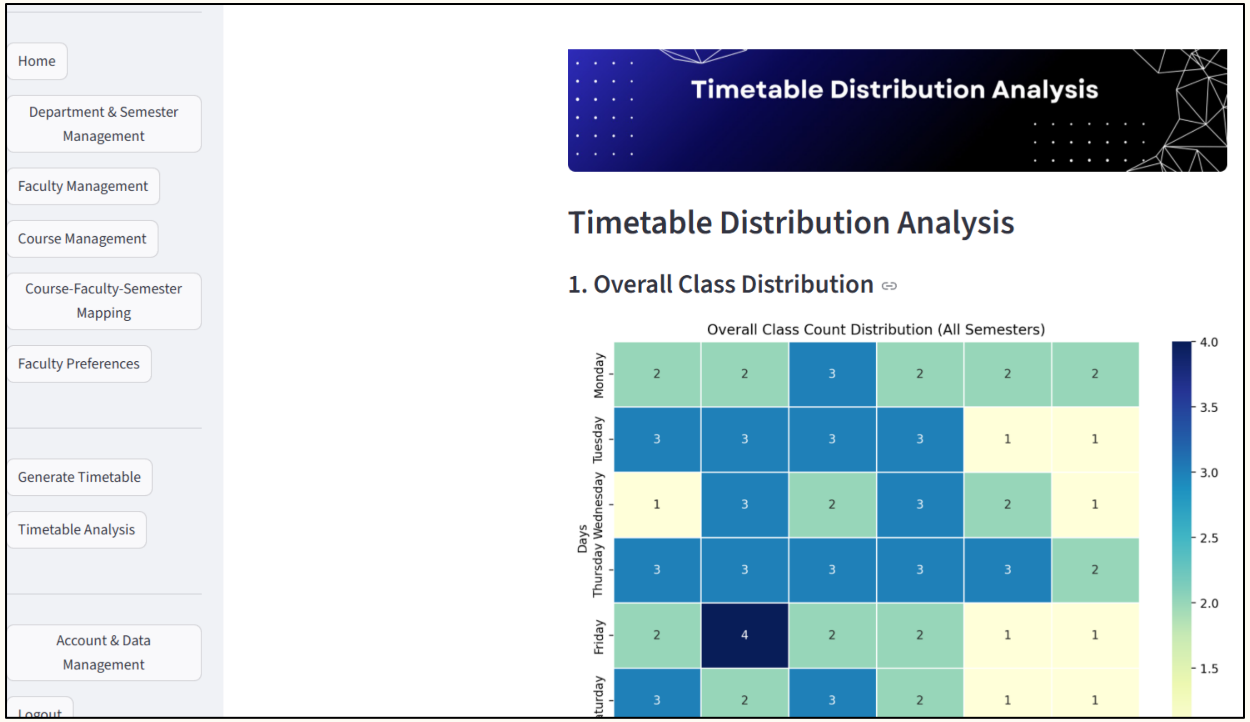
Semester 7

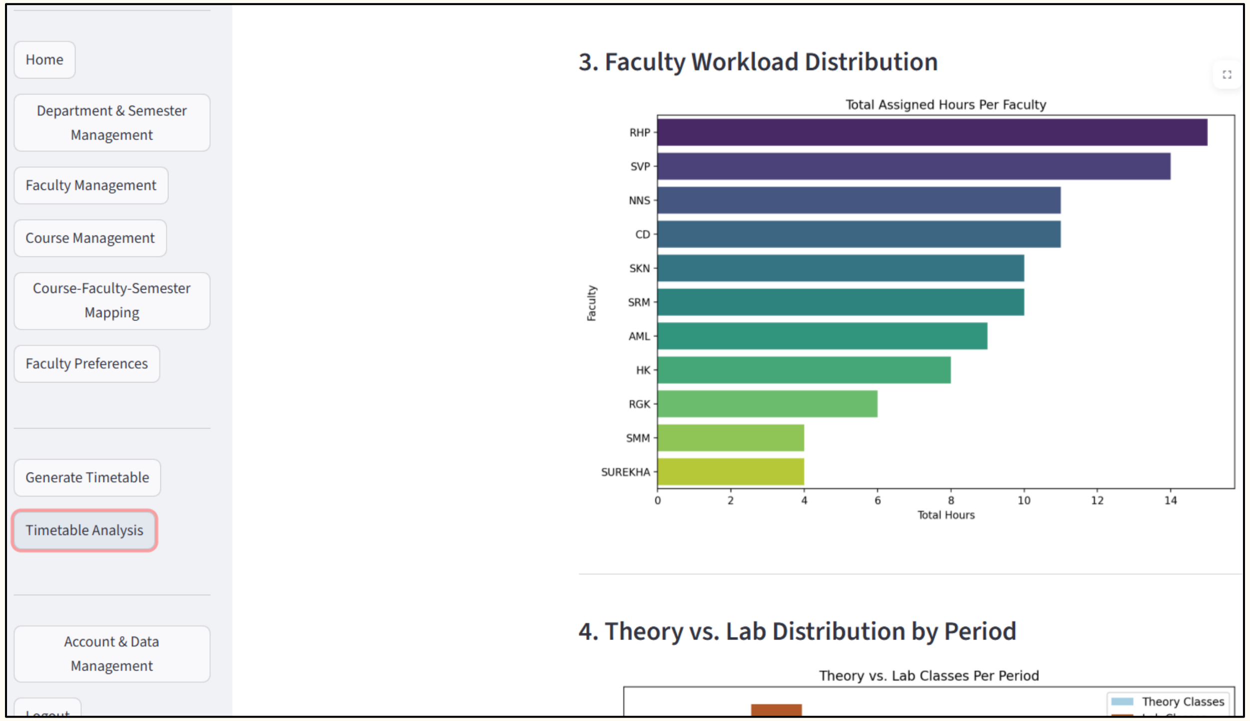
Day	Slot 1	Slot 2	Slot 3	Slot 4	Slot 5	Slot 6
Monday		PP (CD)	SMLDS (RHP)	OPEN ELE (SUREKHA)		
Tuesday	OPEN ELE (SUREKHA)	OPEN ELE (SUREKHA)	BA (SRM)	BA (SRM)		
Wednesday	CNS (NNS)	SMLDS (RHP)	SMLDS (RHP)	PROJ P1 (CD, SRM)	PROJ P1 (CD, SRM)	
Thursday	CNS (NNS)	PP (CD)	PP (CD)	SMLDS LAB (RHP, AML)	SMLDS LAB (RHP, AML)	
Friday		BA (SRM)	CNS (NNS)	CNS (NNS)		
Saturday	PP LAB (CD, HK)	PP LAB (CD, HK)	BA (SRM)	OPEN ELE (SUREKHA)		

Semester 5

Day	Slot 1	Slot 2	Slot 3	Slot 4	Slot 5	Slot 6
Monday	CN (NNS)	TC (RGK)	RMI (SVP)		DV LAB (CD, SVP)	DV LAB (CD, SVP)
Tuesday	TC (RGK)	NOSQL (SKN)	CN (NNS)	SPM (HK)		
Wednesday		CN (NNS)	NOSQL (SKN)	NOSQL (SKN)	MINI PROJ (RHP, SVP)	MINI PROJ (RHP, SVP)
Thursday	NOSQL (SKN)	SPM (HK)	SPM (HK)	TC (RGK)	CN LAB (NNS, HK)	CN LAB (NNS, HK)
Friday	SPM (HK)	NSS (AML) — RMI (SVP)				
Saturday	TC (RGK)	RMI (SVP)	PE (RHP)			

Timetable Visualisation





➢ Conclusion

- The Web-Based Automatic Timetable Scheduler effectively automates the process of academic timetable generation using a Genetic Algorithm. The system produces conflict-free schedules while considering faculty availability and institutional constraints. It reduces manual effort, improves scheduling accuracy, and provides a reliable solution for real-world academic environments.

➢ Genetic Algorithm

- Genetic algorithms are commonly used to generate high-quality solutions to optimization and search problems via biologically inspired operators such as selection, crossover, and mutation.
- **Creating Random Timetables** – The system first creates many random timetable options.
- **Checking for Mistakes** – It examines each timetable to see if there are any clashes, like a teacher being assigned to two classes at the same time.
- **Picking the Best Ones** – The system selects the best schedules that have fewer mistakes.

  “Selects the best, rejects the rest”
- **Mixing and Matching** – It takes the best schedules and combines them to create even better ones.
- **Making Small Changes** – Tiny changes are made to improve the timetable further.
- **Repeating the Process** – This cycle continues until the system finds the most perfect timetable possible.